José Luis Carmona Jiménez

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Employment

Postdoctoral Researcher 16/09/2024–present
Simion Stoilow Institute of Mathematics of the Romanian Academy (IMAR), Bucharest, Romania.
Postdoctoral position supported by the PNRR-III-C9-2023-I8 grant CF 149/31.07.2023, *Conformal Aspects of Geometry and Dynamics*.

Secondary and Upper-Secondary School Teacher 01/09/2022–10/09/2024
Spain.
Permanent position.

Predoctoral Researcher in Differential Geometry 01/10/2020–31/08/2022
Universidad Complutense de Madrid (UCM), Madrid, Spain.
FPI predoctoral fellow in the research group *Geometrical, Topological and Combinatorial Structures Associated with Dynamical Systems*. Principal Investigator: Prof. Marco Castrillón López.

Researcher in Differential Geometry 01/07/2020–30/09/2020
Instituto de Ciencias Matemáticas (ICMAT), Madrid, Spain.
Research grant associated with the research group *Algebra and Geometry*.

Education

Ph.D. in Mathematics 10/05/2024
Universidad Complutense de Madrid, Spain.
Thesis: *Homogeneous descriptions and families of homogeneous structures*. Advisor: Prof. Marco Castrillón López.

M.A. in Secondary Mathematics Education 2019
University of Granada, Spain.

M.Sc. in Advanced Mathematics 2018
Universidad Complutense de Madrid, Spain.
GPA: 9.58/10. Dissertation: *The Ambrose–Singer Theorem*. Advisor: Prof. Marco Castrillón López.

B.Sc. in Mathematics 2017
University of Málaga, Spain.
GPA: 9.17/10.

Research Interests

Differential Geometry; Geometric Analysis; Homogeneous Geometry; Complex Hyperbolic Geometry; Invariant Connections; Conformal Geometry; Weyl Structures; Submanifold Geometry.

Publications

Accepted

[Car26b] José Luis Carmona Jiménez. "On Weyl Structures Reducible in the Direction of the Lee Form". *Proceedings of the American Mathematical Society* (2026). To appear, arXiv: 2507.17496.

[CC25] José Luis Carmona Jiménez and Marco Castrillón López. "Extrinsic Homogeneous Submanifolds". *Note Mat.* **45**:(Suppl. 1) (2025), pp. 117–128. DOI: 10.1285/i15900932v45s1p117.

[CCD25] José Luis Carmona Jiménez, Marco Castrillón López, and José Carlos Díaz-Ramos. "The Ambrose–Singer Theorem for Cohomogeneity One Riemannian Manifolds". *Transformation Groups* (2025). DOI: 10.1007/s00031-025-09927-x.

[CC22a] José Luis Carmona Jiménez and Marco Castrillón López. "The Ambrose-Singer Theorem for General Homogeneous Manifolds with Applications to Symplectic Geometry". *Mediterranean Journal of Mathematics* **19**:(6) (Nov. 2022), p. 280. DOI: 10.1007/s00009-022-02197-x.

[CC22b] José Luis Carmona Jiménez and Marco Castrillón López. "The homogeneous holonomies of complex hyperbolic space". *Annals of Global Analysis and Geometry* **62**:(2) (July 2022), pp. 391–411. DOI: 10.1007/s10455-022-09852-2.

[CC20] José Luis Carmona Jiménez and Marco Castrillón López. "Reduction of Homogeneous pseudo-Kähler Structures by One-Dimensional Fibers". *Axioms* **9**:(3) (2020). DOI: 10.3390/axioms9030094.

Manuscripts in Preparation

[Car26a] José Luis Carmona Jiménez. *On generalized imaginary* Spin^c*-Killing spinors*. In preparation. 2026.

[CCL26] José Luis Carmona Jiménez, Marco Castrillón López, and Eugenia Loiudice. *The Ambrose–Singer Theorem for non-Regular Cohomogeneity One Riemannian Manifolds*. In preparation. 2026.

Teaching Experience

University Teaching

Riemannian Geometry 2023–2024
University of Bucharest, Romania.
Exercise sessions for students in the Master of Science in Mathematics, 6 hours.

Linear Algebra 2020–2022
Universidad Complutense de Madrid, Spain.
Exercise sessions for first-year students in the Faculty of Mathematics and the Faculty of Informatics, 90 hours over two academic years.

Seminar Course: The Ambrose–Singer Theorem 2020
University of Santiago de Compostela, Spain.
Ten-hour seminar course during a research stay at the Faculty of Mathematics.

Secondary Education

Full-time Mathematics Teacher 2022–2024
Spain.
Secondary and upper-secondary teaching at I.E.S. Las Lagunas, Mijas, and I.E.S. Mar de Alborán, Estepona.

Long Research Stays

Philipps-Universität Marburg 15/09/2025–30/11/2025
Marburg, Germany.
Postdoctoral research stay with Dr. Eugenia Loiudice at the Faculty of Mathematics and Informatics.

Philipps-Universität Marburg 01/05/2022–31/07/2022
Marburg, Germany.
Predoctoral research stay with Prof. Dr. habil. Ilka Agricola at the Faculty of Mathematics and Informatics.

University of Santiago de Compostela 20/09/2021–20/12/2021
Santiago de Compostela, Spain.
Predoctoral research stay with Prof. José Carlos Díaz Ramos at the Department of Geometry and Topology.

Short Research Stays

University of Málaga 17/10/2024–28/10/2024
Málaga, Spain.

Short research stay with Prof. Marián Cañadas at the Department of Geometry and Topology. Talk: *On the history of Ambrose–Singer Theorems.*

Philipps-Universität Marburg 23/06/2022–27/06/2022
Marburg, Germany.
Research stay with Dr. Eugenia Loiudice at the Faculty of Mathematics and Informatics.

Research Projects

Geometric Structures on Manifolds, Symmetry and Field Theory 01/09/2025–31/08/2027
Role: Participant.
Principal Investigator: Prof. Marco Castrillón López. Funding entity: Ministry of Science and Innovation, Spain. Funding amount: 91,250.00 EUR. Reference: PID2024-156578NB-I00.

Conformal Aspects of Geometry and Dynamics 01/01/2024–30/06/2026
Role: Participant.
Principal Investigator: Prof. Andrei Moroianu. Funding entity: NextGenerationEU, through the Planul Național de Redresare și Reziliențua, Romania. Funding amount: 6,000,000 RON, approximately 1,200,000 EUR. Reference: PNRR-III-C9-2023-I8 grant CF 149/31.07.2023.

Geometrical, Topological and Combinatorial Structures Associated with Dynamical Systems 01/09/2022–31/08/2025
Role: Participant.
Principal Investigators: Prof. Marco Castrillón López and Prof. Luis Giraldo. Funding entity: Ministry of Science and Innovation, Spain. Funding amount: 108,900.00 EUR. Reference: PID2021-126124NB-I00.

Combinatorics and Dynamics: Geometric and Topological Aspects in Manifolds 01/01/2019–31/12/2021
Role: Participant.
Principal Investigators: Prof. Marco Castrillón López and Prof. Luis Giraldo. Funding entity: Ministry of Science and Innovation, Spain. Funding amount: 68,002.00 EUR. Reference: PGC2018-098321-B-I00.

Algebra and Geometry: Differential Geometry, Symplectic Geometry and Geometric Mechanics 2020
Role: Participant during my research contract at ICMAT.
Reference: 20205-CEX001.

Grants and Fellowships

FPI Predoctoral Fellowship 01/10/2020–31/08/2022
Universidad Complutense de Madrid, Spain.
Funding entity: Ministry of Science and Innovation, Spain. Department: Algebra, Geometry and Topology. Research group: *Geometrical, Topological and Combinatorial Structures Associated with Dynamical Systems.* Principal Investigator: Prof. Marco Castrillón López.

Research Fellowship for Departmental Collaboration 06/11/2017–06/07/2018
Universidad Complutense de Madrid, Spain.
Funding entity: Ministry of Science and Innovation, Spain. Department: Algebra, Geometry and Topology. Advisor: Prof. Marco Castrillón López.

Conference Participation

Conformal Geometry and More – A Conference in Honor of Liviu Ornea July 2025
Bucharest, Romania.
Attendee.

MAM III: Quaternion-Kähler Manifolds and the LeBrun–Salamon Conjecture February 2025
Philipps-Universität Marburg, Germany.
Attendee.

RSME's 7th Congress of Young Researchers January 2025
Bilbao, Spain.
Speaker. Talk: *Symplectic invariant connections.*

The 2nd International Conference on Differential Geometry – ICDG Fez 2024 October 2024
Fez, Morocco.
Speaker. Talk: *Cohomogeneity one invariant connections.*

Conference Differential Geometry and its Applications August 2022
Hradec Králové, Czech Republic.
Speaker. Talk: *The Ambrose–Singer Theorem for symplectic geometry.*

Symmetry and Shape II October 2021
University of Santiago de Compostela, Spain.
Speaker. Talk: *On reductive locally homogeneous Fedosov manifolds.*

Online Summer School on Geometry and Topology August 2021
University of Hradec Králové.
Speaker. Talk: *Fedosov Homogeneous Structures.*

Third BYMAT Conference: Bringing Young Mathematicians Together December 2020
ICMAT, Madrid, Spain.
Speaker. Talk: *The Ambrose–Singer Theorem for general homogeneous manifolds with applications to symplectic geometry.*

Symmetry and Shape I October 2019
University of Santiago de Compostela, Spain.
Poster: *Characterization of simply-connected homogeneous spaces with geometric structures.*

Conference Differential Geometry and its Applications September 2019
Hradec Králové, Czech Republic.
Poster: *A generalization of the Ambrose–Singer Theorem.*

13th ICMAT International Summer School on Geometry, Mechanics and Control July 2019
ICMAT, Madrid, Spain.
Speaker. Talk: *Locally homogeneous Riemannian spaces.*

Second BYMAT Conference: Bringing Young Mathematicians Together May 2019
ICMAT, Madrid, Spain.
Attendee.

12th ICMAT International Summer School on Geometry, Mechanics and Control July 2018
University of Santiago de Compostela, Spain.
Attendee.

Seminars and Short Talks

Geometry Seminar 24/06/2025
Philipps-Universität Marburg, Germany.
Talk: *The Ambrose–Singer Theorem for Cohomogeneity One Riemannian Manifolds.*

Geometry Seminar 11/04/2025
IMAR, Bucharest, Romania.
Talk: *The Ambrose–Singer Theorem for Cohomogeneity One Manifolds.*

Geometry Seminar 21/10/2024
University of Málaga, Spain.
Talk: *On the history of Ambrose–Singer Theorems.*

Geometry Seminar 15/10/2024
IMAR, Bucharest, Romania.
Talk: *Ambrose–Singer Theorems III.*

Geometry Seminar 08/10/2024
IMAR, Bucharest, Romania.
Talk: *Ambrose–Singer Theorems II.*

Geometry Seminar 01/10/2024
IMAR, Bucharest, Romania.
Talk: *Ambrose–Singer Theorems.*

VII Ph.D.ay Mathematics 03/07/2023
Universidad Complutense de Madrid, Spain.
Divulgation talk: *Teselaciones y espacios homogéneos.* English title: *Tessellations and homogeneous spaces.*

IV Conference on Career Opportunities in Mathematics 12/04/2023
Universidad Complutense de Madrid, Spain.
Round table participant.

Geometry and Topology Seminar 04/04/2022
University of Málaga, Spain.
Talk: *The Ambrose–Singer Theorem.*

V Ph.D.ay Mathematics 24/06/2021
Universidad Complutense de Madrid, Spain.
Divulgation talk: *Las geometrías homogéneas del espacio hiperbólico complejo.* English title: *The Homogeneous Geometries of Complex Hyperbolic Space.*

Divulgation Seminar 17/11/2021
University of Santiago de Compostela, Spain.
Talk: *Teselaciones y espacios homogéneos.* English title: *Tessellations and homogeneous spaces.*

III Ph.D.ay Mathematics 02/07/2019
Universidad Complutense de Madrid, Spain.
Divulgation talk: *El teorema de Ambrose–Singer. Un puente entre el Álgebra, la Geometría y el Análisis.* English title: *The Ambrose–Singer Theorem: A Bridge between Algebra, Geometry, and Analysis.*

Languages

Spanish: native. English: advanced.

Technical Skills

- **Computational Mathematics:** Python, NumPy, SymPy, SageMath, symbolic algebra, tensor calculus, Lie algebras.
- **Tools:** LaTeX, Git/GitHub, HTML.